

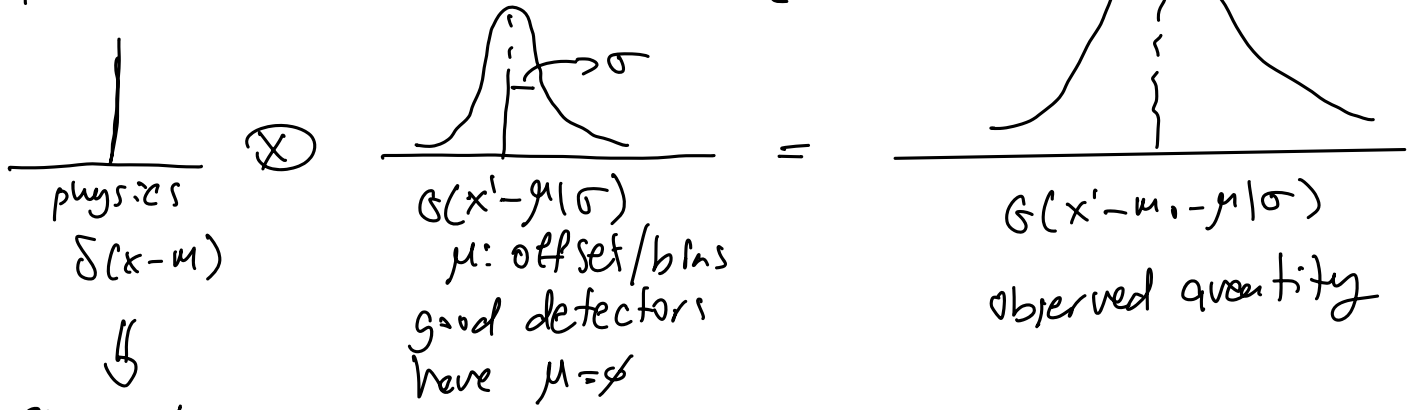
Production of charm quark

1) $e^+e^- \rightarrow X \rightarrow e^+e^-, \mu^+\mu^-$

$X = c\bar{c}$ 

2) $p + \text{target} \rightarrow Xc\bar{c} + \text{pions} + \text{baryons} + \text{kaons} + \dots$
 $\hookrightarrow e^+e^-, \mu^+\mu^-$

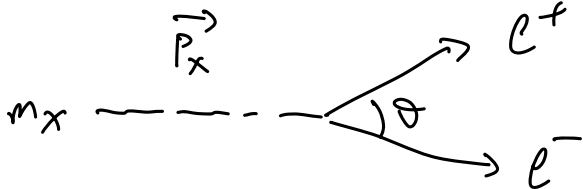
look at invariant mass of $M_{\ell\ell}$



assuming
 intrinsic width Γ
 of particle very small.

$BW(m, \Gamma) = \frac{\Gamma^2}{(x-m)^2 + (\Gamma/2)^2} \xrightarrow{\Gamma \rightarrow 0} \delta(x-m)$

$\vec{P}_X = \vec{P}_{e^+} + \vec{P}_{e^-}$



$m_{\ell\ell}^2 = (\vec{P}_{e^+} + \vec{P}_{e^-})^2 = (E^+ + E^-)^2 - (\vec{P}_{e^+} + \vec{P}_{e^-})^2$

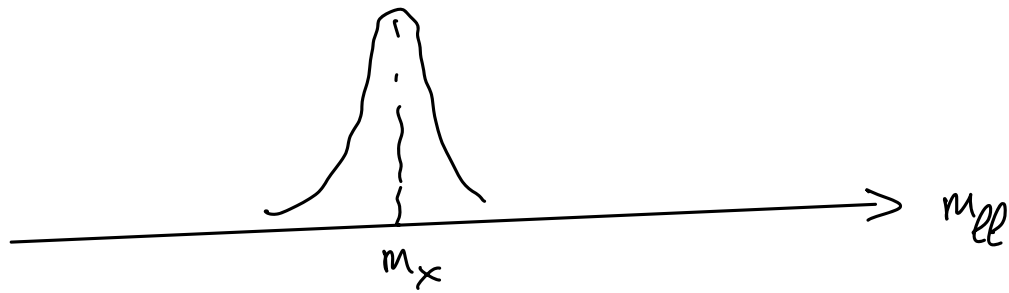
in high energy regime $m_\ell \ll |\vec{P}_\ell|, E_\ell \Rightarrow |\vec{P}_\ell| = E_\ell$

$m_{\ell\ell}^2 = (E^+ + E^-)^2 - (\vec{P}_{e^+}^2 + \vec{P}_{e^-}^2 + 2\vec{P}_{e^+} \cdot \vec{P}_{e^-})$
 $E^{+2} + E^{-2} + 2E^+E^- \cos\theta$

$= 2E^+E^- - 2E^+E^- \cos\theta = 2E^+E^- (1 - \cos\theta) = 4E^+E^- \sin^2 \frac{\theta}{2}$

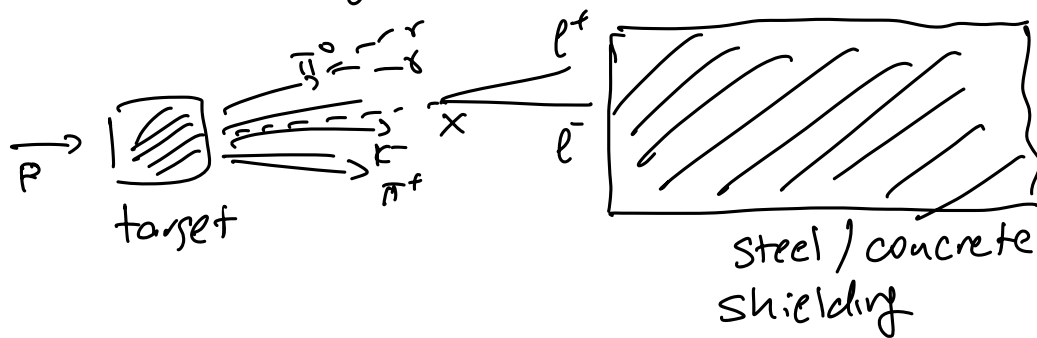
measure $E^+, E^-, \theta \Rightarrow$ Compute $m_{\ell\ell}$

Expected distribution if x exists:



$$\sigma_m^2 = \sigma_E^2 \oplus \sigma_\theta^2$$

σ_E very precise at these energies.



average decay length: $\langle \ell \rangle = \beta \gamma c \tau$

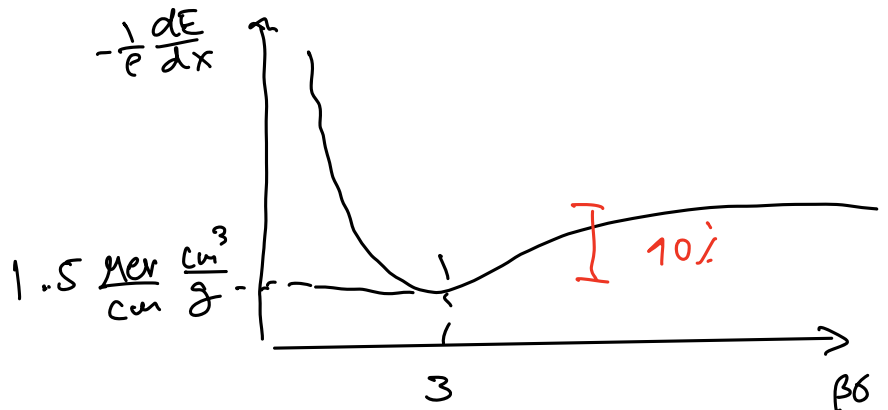
$$\tau \sim 10^{-8} \text{ s} \quad m_\pi \approx 0.1 \text{ GeV} \quad / \quad P = 10 \text{ GeV}$$

$$\beta \gamma = \frac{P}{m} = \frac{10}{0.1} = 100$$

$$\Rightarrow \langle \ell \rangle = 100 \cdot 3 \times 10^8 \text{ m/s} \cdot 10^{-8} \text{ s} \approx 300 \text{ m.}$$

Energy loss = ?

$$\rho = 10 \text{ g/cm}^3$$

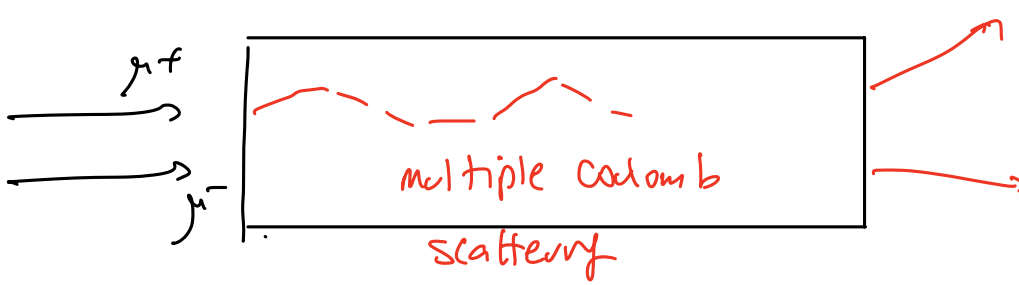


$$\Rightarrow \left\langle \frac{dE}{dx} \right\rangle = 15 \frac{\text{MeV}}{\text{cm}}$$

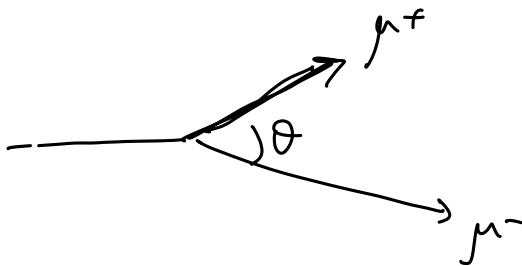
$$\Rightarrow \langle dE \rangle = \left\langle \frac{dE}{dx} \right\rangle \cdot \langle \ell \rangle \Rightarrow 1500 \text{ MeV} = 1.5 \text{ GeV}$$

100m.

For hadrons add nuclear interaction. $\Rightarrow$ stop the hadrons.



$\Rightarrow$ angle of muon momentum modified by material.



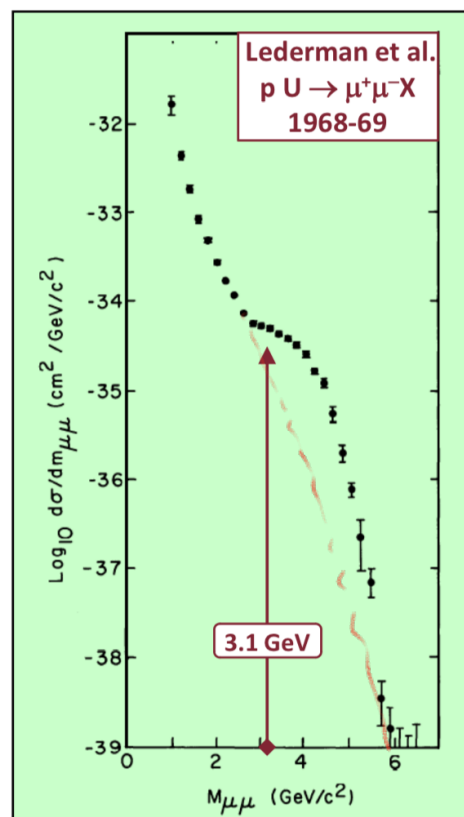
After material.

$$m_{\ell\ell}^2 = 4E^+E^- \sin^2 \frac{\theta}{2}$$

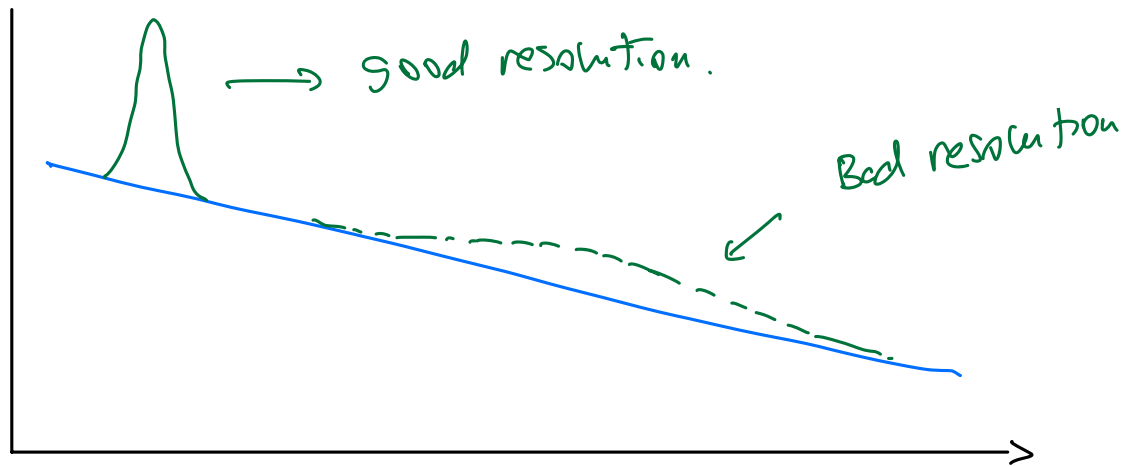
$\Rightarrow$ worse resolution on angle measurement

$$\sigma_m^2 = \sigma_E^2 \oplus \sigma_\theta^2$$

will dominate.

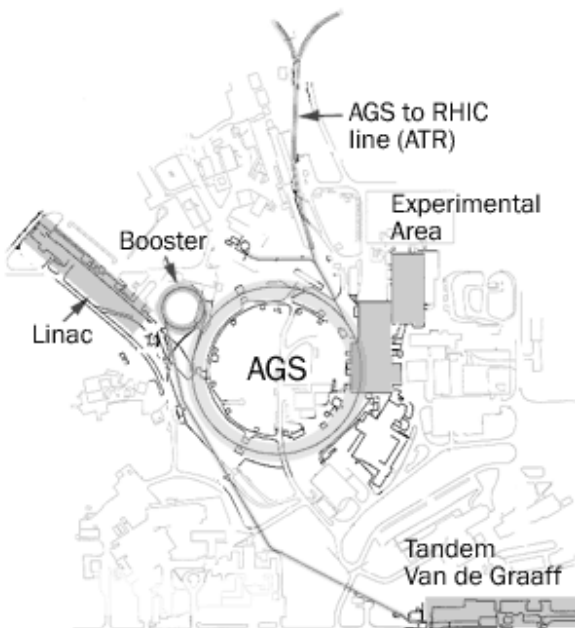


Shoulder in $m_{\mu\mu}$
invariant mass



November Revolution (1974)

- 1) SLAC @ Stanford $e^+e^- \rightarrow X_{c\bar{c}} \rightarrow e^+e^-/\mu^+\mu^-/\text{hadrons}$.
- 2) Brookhaven: $p + \text{Be} \rightarrow X_{c\bar{c}} + \text{hadrons}$
 $\hookrightarrow e^+e^-$ look at m_{ee}



AGS: Alternating Gradient Synchrotron.
 proton beam: $E_{\text{beam}} = 28 \text{ GeV}$

$$\underline{P}_1 = (E_{\text{beam}}, \vec{E}) \quad \bullet (m_p, \vec{0})$$

$$s = (\underline{P}_1 + \underline{P}_2)^2 = m_p^2 + m_p^2 + 2m_p \cdot E_{\text{beam}} \approx 2m_p E_{\text{beam}}$$

$$\Rightarrow \sqrt{s} \approx 7.2 \text{ GeV}$$

$$p + p \rightarrow p + p + X$$

$$m_X \leq \sqrt{s} - 2m_p \approx 5 \text{ GeV}$$

measure Energy and angle
of e^+, e^-

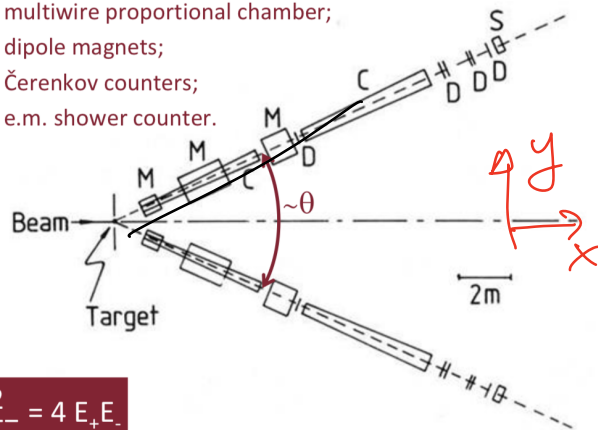
$$M_{ee} = 4 E^+ E^- \sin^2 \frac{\theta}{2}$$

D : multiwire proportional chamber;

M : dipole magnets;

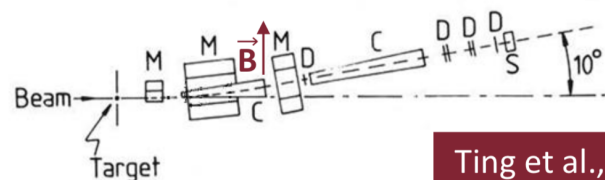
C : Čerenkov counters;

S : e.m. shower counter.



$$m_{+-}^2 = 4 E_+ E_- \sin^2(\theta/2)$$

(a) Plan view



(b) Side view

Ting et al.,
 $p \text{ Be} \rightarrow e^+ e^- X$

$$N_{\text{tot}} = N_S + N_B$$

$$\hat{N}_S = N_{\text{tot}} - N_B$$

significance $\frac{\hat{N}_S}{\sqrt{N_B}}$

$$N_{\text{tot}} \approx 250$$

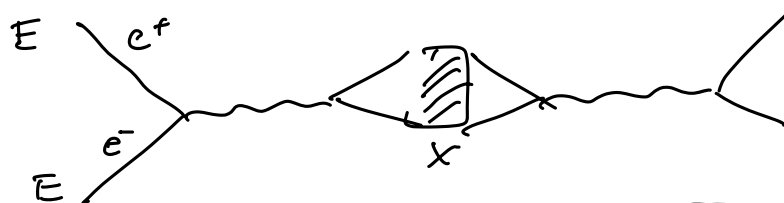
$$N_B \approx 7 \times 10 = 70 \text{ events}$$

$$\frac{N_{\text{tot}} - N_B}{\sqrt{N_B}} = \frac{250 - 70}{\sqrt{70}} = \frac{180}{8.37} \approx 21.5$$

$\Rightarrow$ signal at $m = 3.1 \text{ GeV}$

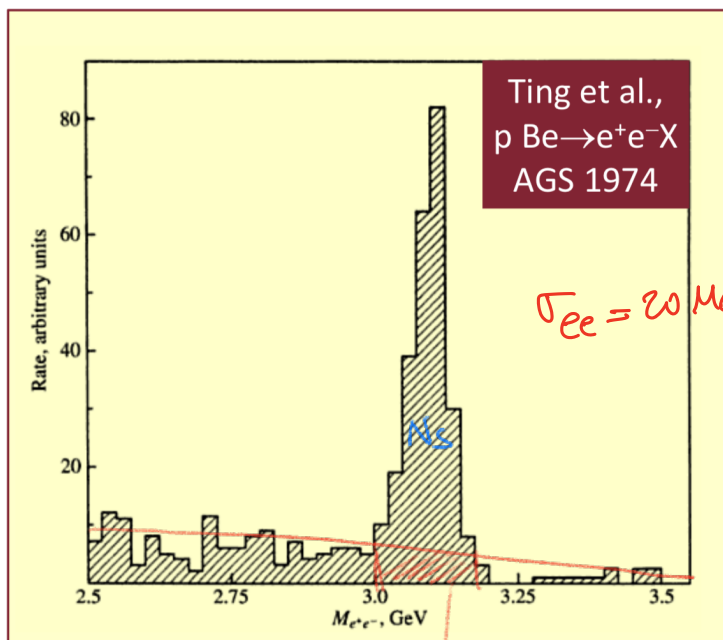
σ_{ee} dominated by exp. resolution $\Rightarrow$ cannot determine the Γ of X

SLAC: $e^+ e^- \rightarrow X \rightarrow e^+ e^-, \mu^+ \mu^-, \text{hadrons.}$



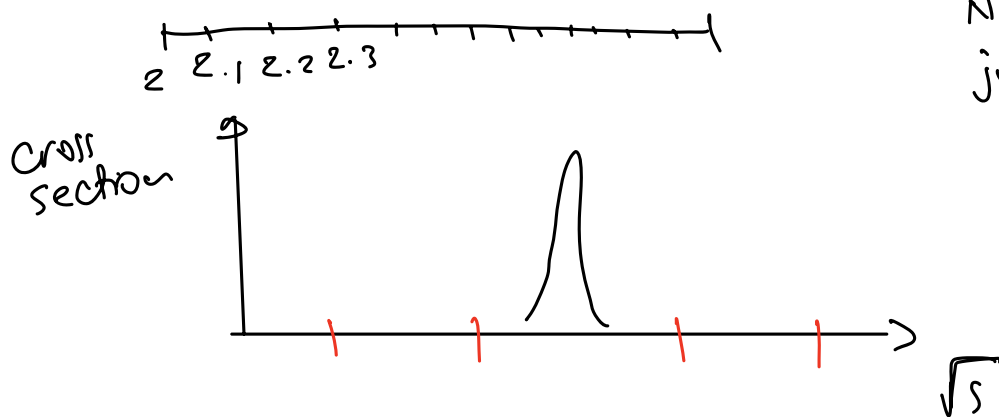
change energy of the beam and scan $\sqrt{s}$

$$\sqrt{s} = 2E$$



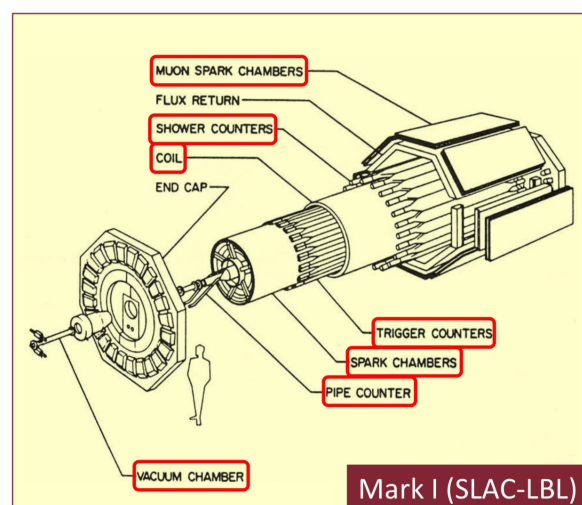
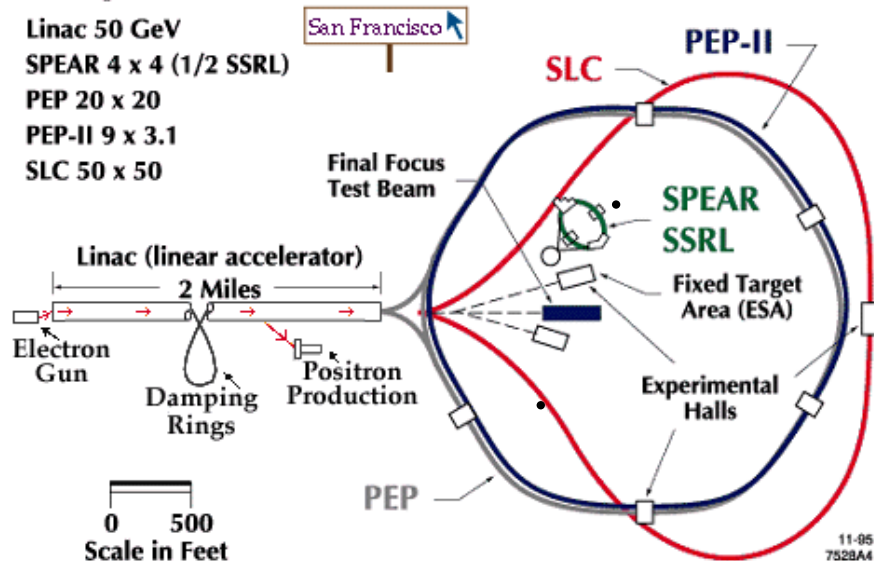
$$\sigma_{ee} = 20 \text{ nb}$$

m_{ee}



Scan near 3.1 with steps of 2.5 MeV

Experimental Areas at SLAC



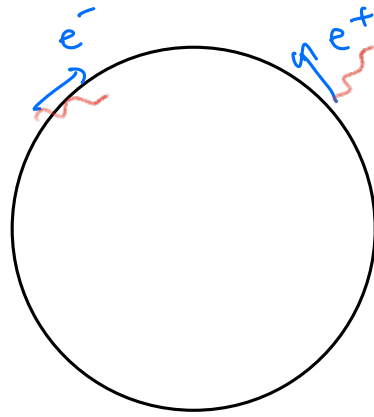
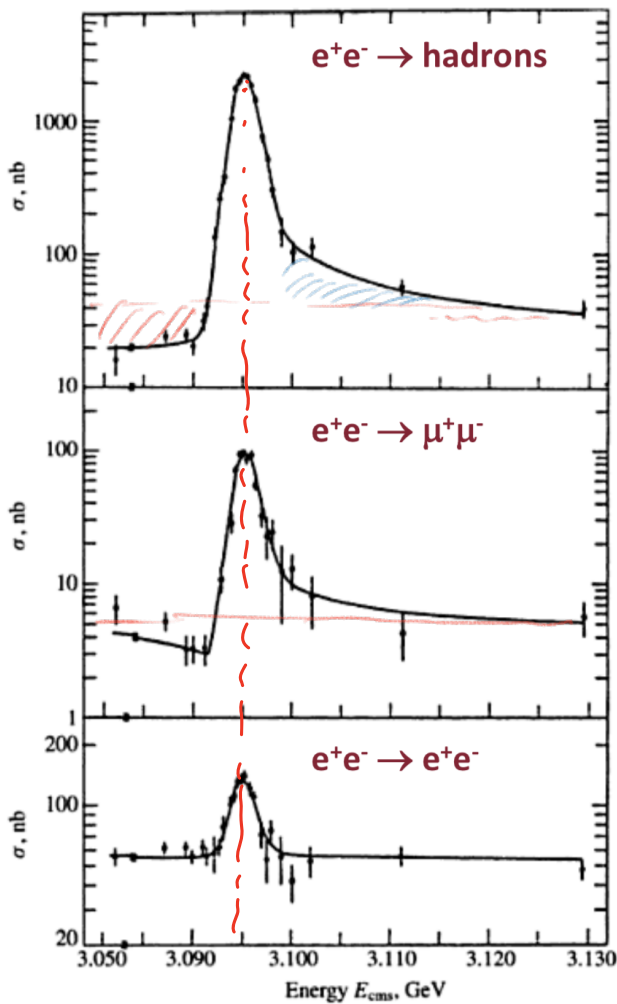
measure # events e^+e^- , $\mu^+\mu^-$, hadrons for different $\sqrt{s} = 2E_{beam}$.

$$\sigma(e^+e^- \rightarrow X \rightarrow f\bar{f}) = \frac{16\pi}{s} \frac{2J_X + 1}{(2S_1 + 1)(2S_2 + 1)} \frac{\Gamma_{ee}}{\Gamma_X} \frac{\Gamma_{f\bar{f}}}{\Gamma_X} \frac{\Gamma_X^2}{(m_X - \sqrt{s})^2 + (\Gamma_X/2)^2}$$

$S_1 = S_2 = 1/2$
for e^+e^-

Γ_X : total width of X particle

J_X : total J of X



$$\sqrt{s} = E^+ + E^- \quad (\text{nominal}) \quad \sqrt{s}$$

$$\sqrt{s}_{\text{nom}} = 2 E_{\text{beam}}$$

$$E^+ = E^{+'} + E_{\gamma} \quad \rightarrow \text{emission.}$$

$$\sqrt{s} = E^{+'} + E^- = \sqrt{s}_{\text{nom}} - E_{\gamma}$$

$$\sigma \text{ Max for } \sqrt{s}_{\text{nom}} = m_X$$

$$\sqrt{s}_{\text{nom}} = 3050 \text{ MeV} \quad \text{below the peak.}$$

$$\sqrt{s}_{\text{eff}} < \sqrt{s}_{\text{nom}} \quad \text{because of radiation of } \gamma$$

$$\Rightarrow \sigma_{\text{eff}} < \sigma_{\text{nom}} \quad \sigma \text{ decreases because of radiation.}$$

$$\sqrt{s}_{\text{nom}} = 3150 \text{ MeV}$$

$$\text{Radiation} \Rightarrow \sqrt{s}_{\text{eff}} < \sqrt{s}_{\text{nom}} \Rightarrow \text{get closer to peak}$$

$$\Rightarrow \sigma(\sqrt{s}_{\text{eff}}) > \sigma(\sqrt{s}_{\text{nom}})$$

$$\text{measure the same mass } m = 3010 \text{ MeV}$$

Ting: J particle

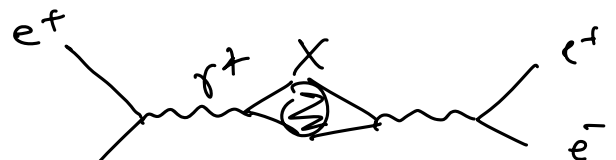
J/psi in PDG.

Richter: psi

$$J^{PC} = ? \quad \text{for } J/\psi$$

J must be the same as gamma

$$\Rightarrow J = 1.$$



$$C = (-1)^{L+S}$$

$$P = (-1)^{L+1}$$

$$\text{ground state} \Rightarrow L=0. \Rightarrow P=-1 \Rightarrow C = (-1)^{0+1} = -1$$

$$L=0 \Rightarrow S=1$$

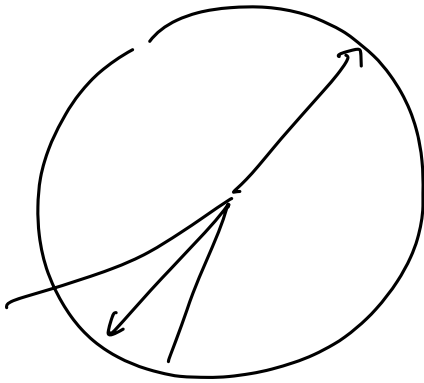
$$J/\psi \quad J^{PC} = 1^{--}$$

vector meson of $c\bar{c}$

Fucreas Elser

$\Rightarrow$ look for new resonance.

$\Rightarrow$ 10 days later $\psi' \equiv \psi(2S)$



$$M_{\mu\mu} = 2E^+ E' (1 - \cos\theta)$$

